9. Empirical Confirmations?

The appeal of quantum cosmology goes beyond the fascinating theoretical explorations of what there might be beyond the Big Bang. There is another reason for studying the application of the theory to cosmology: it might provide the opportunity to find out whether or not the theory is actually correct.

Science works because, after hypotheses and reasoning, after intuitions and visions, after equations and calculations, we can check whether we have done well or not: the theory gives predictions about things we have not yet observed, and we can check whether these are correct, or not. This is the power of science, that which grounds its reliability and allows us to trust in it with confidence: we can check whether a theory is right or wrong. This is what distinguishes science from other kinds of thinking, where deciding who is right and who is wrong is usually a much thornier question, sometimes even devoid of meaning.

When Lemaître defends the idea that the universe is expanding, and Einstein does not believe it, one of the two is wrong; the other right. All of Einstein's results, his fame, his influence on the scientific world, his immense authority, count for nothing. The observations prove him wrong, and it's game over. An obscure Belgian priest is right. It is for this reason that scientific thinking has power.

The sociology of science has shed light on the complexity of the process of scientific understanding; like any other human endeavour, this process is beset by irrationality, intersects with the game of power and is affected by every sort of social and cultural influence. Nevertheless, despite all of this, and in opposition to the exaggerations of a few postmodernists, cultural relativists and the like, none of this diminishes the practical and theoretical efficacy of scientific thinking. Because in the end, in the majority of cases, it is possible to establish with clarity who is right and who is wrong. And even the great Einstein could go on to say (and he did so), 'Ah … I made a mistake!' Science is the best strategy if we value reliability.

This does not mean that science is just the art of making measurable predictions. Some philosophers of science overly circumscribe science by limiting it to its numerical predictions. They miss the point, because they confuse the instruments with the objectives. Verifiable quantitative